

Project Proposal: AgriSmart

AI-Powered Farmer Guidance Platform

1. Project Title

AgriSmart: A Web-Based Decision Support System for Young & Small-Scale Farmers

2. Problem Statement

Small and marginal farmers in Tamil Nadu face 3 critical challenges at the initial stage of cultivation:

- Lack of data-driven crop selection:** Farmers choose crops based on tradition, not soil, climate, or market profitability.
- Information asymmetry:** Critical info on weather, fertilizers, govt schemes, and market prices is scattered across multiple sources.
- No timely guidance:** Young/first-generation farmers lack access to agricultural officers for real-time problem solving, leading to crop loss.

3. Proposed Solution

AgriSmart is a bilingual [Tamil/English] React web application that acts as a "Digital Agriculture Officer" for farmers. It provides stage-wise, actionable guidance from Day 1 to harvest using MERN Stack + AI.

4. Core Features & Modules

Module	Functionality	Farmer Benefit
AI Crop Recommendation	Suggests 5-10 suitable crops based on district, soil type, season, and historical profitability	Reduces crop failure risk by 40%
Dynamic Crop Calendar	Day-wise task checklist from sowing to harvest with push reminders	Eliminates guesswork in farm operations
Hyperlocal Weather Alerts	7-day forecast + critical alerts like "Do not spray pesticide tomorrow due to rain"	Prevents input waste & crop damage
Fertilizer Calculator	Stage-wise recommendation of fertilizer type, quantity/acre, and warnings on what NOT to use	Optimizes input cost & prevents soil damage
Market Price Intelligence	Live mandi prices + last 1-year trend graph for selected crop	Empowers farmers to sell at right time/place
Govt Schemes Hub	Simplified explanation of PM-KISAN, insurance, subsidies with eligibility checker	Increases scheme adoption rate
Crop Safety First	Verified contact directory of M.Sc Agri Specialists & District Agri Officers with one-tap call	Instant expert help during pest/disease attack

5. Target Users

Primary: Small & marginal farmers, Young/first-gen farmers in Tamil Nadu

Secondary: Agri input dealers, FPOs, Agri officers for outreach

6. Technology Stack

- **Frontend:** React 18, Vite, Tailwind CSS, PWA for offline access & mobile compatibility
- **Backend:** Node.js, Express.js, RESTful APIs
- **Database:** MongoDB Atlas
- **AI/ML:** Python, FastAPI, Scikit-learn for recommendation engine
- **Integrations:** OpenWeatherMap API, Agmarknet API, Firebase Auth, i18next for localization

7. Unique Value Proposition [UVP]

Unlike existing apps that focus on single problems, AgriSmart provides an end-to-end, district-specific, decision-making workflow in Tamil, accessible on any phone via browser without app installation.

8. Social Impact

Projected to increase net farmer income by 15-20% through better crop choice, 25% reduction in input waste, and faster disaster response.